# Fast SAR Image Segmentation With Deep Task-Specific Superpixel Sampling and Soft Graph Convolution

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***Abstract*— Since the number of superpixels is lower than that of pixels, superpixels can substantially speed up subsequent processing steps and have been widely used in synthetic aperture radar (SAR) image segmentation. However, in most of the existing superpixel-wise segmentation algorithms, superpixel prediction is an isolated preprocessing step and is independent of the segmentation task. The performance of the segmentation results is determined by the accuracy of superpixels. Once superpixels are generated, their shape cannot be changed in the following segmentation stage, even if the same superpixels contain pixels of different landcovers. To address this, we propose an end-to-end trainable superpixel-wise segmentation method for single-polarization SAR images. First, we design a differentiable boundary-ware clustering method for estimating task-specific superpixels. Instead of the hard association between pixels and superpixels in the existing superpixel algorithms, this method introduces the soft association map to make the clustering differentiable. Hence, it can be implemented using a simple deep fully convolutional network. In the segmentation part, we propose a novel soft graph convolution network (SoftGCN), which takes the association map as input and performs superpixel-wise segmentation. The advantage of our method is that superpixel generation and graph convolution parts can be trained under a unified framework, until two parts obtain the optimum parameters. In the training process, it can adaptively adjust the shape of the superpixels according to the segmentation results, ensuring the superpixels correctly adhere the boundaries. Experimental results with simulated and real SAR images demonstrate that our method outperforms other state-of-the-art segmentation algorithms, while also being faster.**

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## I. Introduction

SYNTHETIC aperture radar (SAR) can provide high-resolution images in all-weather and day-night conditions. SAR segmentation, aiming at assigning a certain label to every pixel, is the foundation for many applications of SAR (e.g., change detection [1], target recognition [2], [3].)

The common segmentation frameworks are pixel level, which first adopts gray-level co-occurrence matrix (GLCM) [4] or transforms domain filters to extract the features of SAR images, such as wavelet transform [5] and Gabor filter [6]. Then, the classifiers are applied to label the pixels with different categories, including Random Forest (RF) [7], AdaBoost [8], Naive Bayesian [9], and so on. Recently, deep learning architectures have been proved to own excellent high-level feature representation performance. Moreover, they can perform feature extraction and segmentation at the same time. Until now, the deep methods based on the convolutional neural networks (CNNs) [10] or recurrent neural networks (RNNs) [11] have achieved excellent performance dealing with several popular SAR datasets [12]. However, all the above algorithms are on pixel level, whose computational burden is positively correlated with the number of pixels. The pixel-level algorithms always face the intensive computation burden in practical application. Particularly, the development of SAR imaging sensors vastly enhances the resolution of images.

Superpixel is a local, homogeneous region in which pixels have similar attributes [13]. Since the number of superpixels is far less than that of pixels, superpixel-wise segmentation methods usually have better computational efficiency and have received considerable attention. These algorithms generally include three stages. First, the images are over-segmented into superpixels with both high boundary adherence and visual compactness. For example, Xiang *et al.* [14] put forward an adaptive single-polarization SAR superpixel generation approach based on linear feature clustering and edge constraint. Zou *et al.* [15] used generalized Gamma distribution to model accurately a variety of high-resolution SAR scenes and over-segment the

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SAR images into superpixels. Akyilmaz and Leloglu [16], Qin *et al.* [17], and Song *et al.* [18] adopted the Mahalanobis distance (MD) [19], revised Wishart distance [20], and Bartlett distance, respectively, to replace the Euclidian distance to measure the similarity between pixels, achieving good performance on SAR superpixel generation. Second, the superpixel-wise feature vectors are extracted considering both the speckle noise and complex geometric distortions within SAR images. Finally, the supervised or unsupervised clustering algorithms are introduced to obtain the superpixel-level classification results. For example, Xiang *et al* [21] proposed a semantic superpixel-based classification integrating the superpixels and high-level semantic features. Pappas *et al.* [22] used agglomerative hierarchical clustering to group the superpixels into multiple clusters representing different landcover types. Some algorithms also used probabilistic graphical models to conduct superpixel-wise segmentation. One of the most representative models is conditional random fields (CRFs). CRF regards each superpixel as a node of graph model. However, the inference process of probabilistic graphical models usually has heavy computational burden.

Inspired by the great breakthrough of deep learning in SAR image interpretation, some researchers attempted to modify the deep networks to incorporate SAR image segmentation. For example, in the superpixel generation part, Jampani *et al.* [23] proposed a differentiable clustering algorithm for superpixel segmentation and implemented it using deep networks. For the superpixel-wise feature extraction steps, the previous methods are all based on hand-crafted characteristics, which largely depend on professional expertise. Ma *et al.* [24] put each superpixel in a larger patch, where the pixels not covered by superpixels were zero. So that CNN could be used to directly extract superpixels' feature vectors. For superpixel-level segmentation stage, as the irregular superpixel lattice set belongs to the non-Euclidean structure data, Geng *et al.* [25] proposed an attention graph convolution network (GCN) to predict the labels of superpixels.

It is worth pointing out that although these improvements have obtained better segmentation accuracy and higher computational efficiency to a certain extent, they still regard the superpixel generation, feature extraction, and superpixel-wise segmentation as three independent parts. Once the superpixels are generated, their shapes will not change with the segmentation tasks. The performance of segmentation highly relies on the quality of superpixel generation and discrimination of task-specific feature extraction. It is difficult for these three parts to achieve the best performance at the same time. Therefore, to make the performance of segmentation method have a big improvement, superpixel generation, feature extraction, and segmentation stages should be trained under a unified framework, ensuring that they capture the optimal parameters at the same time.

To address the above problems, we propose a superpixel-wise segmentation network for single-polarization SAR images based on deep task-specific superpixel sampling and graph convolution (DSSG) in this article. First, we use local gradient ratio pattern (LGRP) to extract the features, which shows good robustness to speckle noise within SAR images. Second, a differentiable boundary-ware clustering method is proposed to over-segment the SAR image into superpixels. Since this method is differentiable, it can be implemented using a simple fully convolutional network. Moreover, we put forward boundary-ware loss function to suppress the interference from multiplicative speckle noise and data imbalance issue. The superpixel generation network outputs a soft association map, whose elements are continuous variables. In the segmentation stage, we propose a soft graph convolution network (Soft-GCN) to perform superpixel-wise SAR segmentation. The superpixel generation and graph convolution parts can be trained simultaneously under a unified framework to capture the global optimum solution. Experiments show that this method's superpixels have higher boundary adherence and better segmentation performance at the same time.

Overall, our superpixel-wise segmentation method has made the following innovations.

1) We propose a differentiable boundary-ware clustering method that leverages deep networks for estimating task-specific superpixels. This method introduces the soft association maps to replace the hard association matrix in the existing clustering-based superpixel algorithms.
2) A Soft-GCN is proposed, which can directly transform the soft association map into a graph model and perform superpixel-wise segmentation.
3) The deep superpixel sampling network and Soft-GCN constitute an end-to-end trainable superpixel-wise SAR segmentation method. Superpixel prediction and superpixel-wise classification are implemented under a unified deep network framework and can be trained simultaneously according to the task-specific loss functions until they achieve the best segmentation results.

The rest of this article is organized as follows. The existing superpixel generation algorithms are reviewed in Section II. Section III presents the proposed method, including the LGRP feature extraction, deep SAR superpixel prediction network, boundary-aware reconstruction loss, and superpixel segmentation with graph convolution. The experimental results and discussion are presented in Section IV. Section V presents the conclusion.

## II. Superpixel Generation for SAR Images

The existing SAR superpixel algorithms can be roughly divided into clustering-based and graph-based methods. The clustering-based method adopts conventional clustering algorithm to generate superpixel. For example, linear spectral clustering superpixel (LSC) [26], simple linear iterative clustering (SLIC) [27], and manifold SLIC [28] all leverage $k$-means to group the neighbor pixels into superpixels. SLIC uses position and Lab color (5-D) of each pixel as its feature representation, whereas LSC further projects these feature vectors into a 10-D space to have a better clustering performance. Manifold-SLIC represents each pixel as a 2-D manifold feature to produce content-sensitive superpixels. The superpixels using non-iterative clustering (SNIC) method [29] leverages a non-iterative clustering framework to cluster pixels.

In the graph-based approaches, the superpixel generation process is described as a graph partitioning problem, in which nodes represent the pixels and edges mean the pixels are

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adjacent. Essentially, graph partitioning problem is a discrete optimization problem. The common algorithms are normalized cuts [30], Felzenszwalb and Huttenlocher (FH) [31], and entropy rate superpixels (ERs) [13]. Since the variables are discrete, optimization of graph-based methods is usually non-differentiable, meaning that it cannot be implemented by deep networks.

Beyond that, recently some other image processing algorithms are used to generate superpixel [32], such as watershed transform [33], mean shift [34], graph-cuts [35], and geometric flows [36], hill-climbing [37]. But only a few efforts have been made to incorporate them into deep neural networks. It is largely because the existing superpixel algorithms are non-differentiable. Recently, Jampani *et al.* [23] revised the iterative $k$-means clustering algorithm to make it differentiable and proposed an end-to-end deep superpixel estimating networks. In this framework, CNN is used as a feature extractor. Then, a differentiable $k$-means module clusters the adjacent and similar pixels into the same superpixels based on these features. Yang *et al.* [38] further improved the $k$-means module so that the whole superpixel estimating procedure can be implemented using a simple fully convolutional network. Experiments showed that it consistently outperforms the state-of-the-art method with low computational complexity. However, these algorithms are originally designed for natural images, regardless of the multiplicative speckle noise [39] within SAR images in SAR images. Their generation results for SAR images are unsatisfactory in some cases.

## III. Methodology

We will introduce our DSSG method in detail in this section. The first part uses the LGRP feature vectors to replace the original pixel values. The second part proposes the deep SAR superpixel estimating network. The four subsection explains how the boundary-ware loss function suppresses the effect of speckle noise and gradient variations in SAR images. In the last subsection, we introduce the Soft-GCN to perform superpixel-level SAR image segmentation.

### A. LGRP Feature

The coherent imaging property leads to inherent multiplicative speckle noise in SAR images. In addition, the SAR imaging system is more sensitive to gradient variation caused by interaction of signature with the environment, shadowing effects, and the backscattering of the target and background. Both make the SAR image segmentation more difficult than natural images.

In this part, we introduce a highly efficient SAR image feature extraction model LGRP before superpixel generation. LGRP is previously proposed by us for SAR target recognition [40], which improves the local binary pattern (LBP) according to the particularity of SAR images. It has been proven to be very robust to multiplicative speckle noise and gradient changes in SAR images, ensuring the high accuracy of subsequent superpixel generation and classification [41].

LBP is a good image texture feature descriptor with low computational complexity and rotation invariance, widely used in image classification. LBP representor is calculated as follows:

$$\mathrm{LBP}_{P,R} = \sum_{p=0}^{P-1} s\left(g_p - g_c\right)2^p, \quad s(x) = \left\{ \begin{array}{ll} 1, & x \geq 0 \\ 0, & x < 0 \end{array} \right\} \tag{1}$$

where $g_c$ and $g_p$ are the values of center and neighbor pixels, respectively. $P$ is the number of neighbor pixels and $R$ is the radius of the sliding window. $s\ (\cdot)$ is a sign function. Essentially, LBP feature is the statistics of the Euclidean distance between the central pixel and the neighbor pixels in the sliding window.

Considering the interference of multiplicative speckle in SAR image, the LGRP feature of center pixel is defined as

$$\mathrm{LGRP}_{P,R}(g_c) = \frac{1}{P}\sum_{p=1}^{P} s\left(G_{\mathrm{ratio}}\left(g_p\right) - G_{\mathrm{ratio}}(g_c)\right)2^P \tag{2}$$

where $G_{\mathrm{ratio}}(g_p)$ and $G_{\mathrm{ratio}}(g_c)$ are the gradient ratio pattern of the neighbor pixels $g_p$ and center pixel $g_c$, respectively, as shown in the following equations:

$$G_{\mathrm{ratio}}\left(g_p\right) = \frac{|g_p - g_c|}{g_p} \tag{3}$$

$$G_{\mathrm{ratio}}(g_c) = \frac{1}{P}\sum_{p=1}^{P} G_{\mathrm{ratio}}\left(g_p\right). \tag{4}$$

$G_{\mathrm{ratio}}(g_p)$ is defined as the ratio of its gradient value and gray value, whereas $G_{\mathrm{ratio}}(g_c)$ is the mean of $G_{\mathrm{ratio}}(g_p)$. Compared with LBP, the pixels are represented using gradient ratio patterns instead of gray values in LGRP. On one hand, the pixel ratio measurement in LGRP has been demonstrated to have good robustness to multiplicative speckle noise [21], [40]. On the other hand, as the intensities of neighbor pixels and the gradient information in each sliding window are both considered, LGRP is insensitive to the common gradient variations in SAR images with complex scenes.

### B. Deep SAR Superpixel Generation Network

Given an SAR image with size $H \times W$, denoted as $\mathbf{I} \propto \mathbb{R}^{H\times W}$, each pixel has an $R$-dimensional LGRP feature vector. The existing superpixel generation methods' tasks are to compute a hard pixel-superpixel association map $\mathbf{Q} \propto \mathbb{R}^{N\times M}$, assigning each pixel to one superpixel. Here, $N$ and $M$ are the numbers of pixels and superpixels, respectively. $q_{i,j}^p$ denotes an element of $\mathbf{Q}$ and $q_{i,j}^p = 1$ means that pixel $(i, j)$ is assigned to superpixel $p$, and 0 otherwise.

The traditional clustering-based superpixel generation methods, e.g., SLIC, generally have two steps. For pixel $(i, j)$, we first calculate the Euclidean distance between it and all superpixels in the feature space. Then, this pixel will be associated with the nearest superpixel $m$, that is, $q_{i,j}^m = 1$. Subsequently, the average pixel feature inside each superpixel is used to replace the superpixel's original feature. These two steps are repeated until either convergence or for a fixed number of iterations. Note that this nearest neighbor operation is non-differentiable, so they cannot be implemented using deep networks. Jampani *et al.* [23] proposed a differentiable SLIC method for optical images, making it possible to integrate

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superpixel segmentation into trainable deep neural networks. This method first uses the CNN as a means to extract pixel features, which are fed to a soft iterative $k$-means clustering module to compute association map. However, it still has heavy computation cost and is hardly integrated into other tasks. Moreover, it does not take account of the multiplicative speckle noise and gradient variation in SAR image, so directly applying it to SAR images cannot achieve satisfactory results.

In this section, we propose a superpixel generation method for SAR images. We first partition the SAR image $\mathbf{I} \propto \mathbb{R}^{H\times W}$ into $M$ grid cells with a size of $h \times w$, which are regarded as initial superpixels. Instead of calculating the hard association matrix in non-differentiable nearest-neighbor operation, we use an encoder–decoder deep network to compute a soft association $\mathbf{Q} \propto \mathbb{R}^{H\times W\times M}$. $q_{i,j}^m \in [0, 1]$ represents an element of $\mathbf{Q}$, whose coordinate in $\mathbf{Q}$ is $(i, j, m)$. $q_{i,j}^m$ denotes the probability of assigning pixel $(i, j)$ to the $m$th superpixel. The closer $q_{i,j}^m$ is to one, the more likely pixel $(i, j)$ belongs to the $m$th superpixel.

The network is a standard encoder–decoder architecture with skip connections to predict superpixel association map $\mathbf{Q}$. Let $f_{i,j} \propto \mathbb{R}^{1\times R}$ be the $R$-dimensional LGRP feature vector of pixel $(i, j)$, and $p_{i,j} \propto \mathbb{R}^{1\times 2}$ is the pixel's coordinates (2-D). The encoder takes the R + 2-D LGRP position feature maps $\Phi \propto \mathbb{R}^{H\times W\times(R+2)}$ as the input. Then, the encoder network extracts high-level semantic information using a plurality of convolution layers. The following decoder network involves multiply deconvolution layers. It can convert feature maps into soft association maps $\mathbf{Q}$, without computing the distance between pixels and superpixels. Then, we can get the superpixel centers as follows:

$$\hat{f}_m = \frac{\sum_{(i,j)\in\mathbf{I}} f_{i,j}\cdot q_{i,j}^m}{\sum_{(i,j)\in\mathbf{I}} q_{i,j}^m} \tag{5}$$

$$\hat{p}_m = \frac{\sum_{(i,j)\in\mathbf{I}} p_{i,j}\cdot q_{i,j}^m}{\sum_{(i,j)\in\mathbf{I}} q_{i,j}^m} \tag{6}$$

where $\hat{f}_m \propto \mathbb{R}^{1\times R}$ and $\hat{p}_m \propto \mathbb{R}^{1\times 2}$ are the LGRP feature and location vector of superpixel $m$, respectively. $\sum_{(i,j)\in\mathbf{I}} q_{i,j}^m$ is the normalization constant. Mathematically, $\hat{f}_m$ and $\hat{p}_m$ are the weighted sum of all pixels' features and locations. For convenience, we can normalize matrix $\mathbf{Q}$ to make $\sum_{(i,j)\in\mathbf{I}} q_{i,j}^m = 1$, and thus, we can rewrite the above superpixel center update process as

$$\begin{bmatrix} \hat{f}_1 & \hat{p}_1 \\ \vdots & \vdots \\ \hat{f}_M & \hat{p}_M \end{bmatrix} = \tilde{\mathbf{Q}}^{\mathrm{T}}\Phi \tag{7}$$

where $\tilde{\mathbf{Q}}$ is the normalized $\mathbf{Q}$, and T denotes the matrix transpose operation.

In practice, it is time-consuming and unnecessary to compute all superpixels. We constraint its search region to the finite surrounding grid cells. As illustrated in Fig. 1, a pixel in the yellow box can only be assigned to the nine neighbor grid cells (red box), denoted as $\mathcal{N}_{i,j}$. Consequently, the size of $\mathbf{Q}$ can be simplified from $H \times W \times M$ to $H \times W \times |\mathcal{N}_{i,j}|$ where $|\mathcal{N}_{i,j}| = 9$. This approximation in the $\mathbf{Q}$ computation




Fig. 1. Illustration of $\mathcal{N}_{i,j}$. For every pixel (green circle) in the yellow box, only the surrounding nine grid cells in the red box are considered when we compute the pixel-superpixel associations.

can efficiently reduce the requirement for machine's memory and computing power.

Then, given $\mathbf{Q}$ and new superpixel centers, we can reconstruct the LGRP feature vector and the location of each pixel as follows:

$$\hat{f}_{i,j} = \frac{\sum_{m=1}^{M}\left(\hat{f}_m\cdot q_{i,j}^m\right)}{\sum_{m=1}^{M} q_{i,j}^m} = \frac{\sum_{m\in\mathcal{N}_{i,j}}\left(\hat{f}_m\cdot q_{i,j}^m\right)}{\sum_{m\in\mathcal{N}_{i,j}} q_{i,j}^m} \tag{8}$$

$$\hat{p}_{i,j} = \frac{\sum_{m=1}^{M}\left(\hat{p}_m\cdot q_{i,j}^m\right)}{\sum_{m=1}^{M} q_{i,j}^m} = \frac{\sum_{m\in\mathcal{N}_{i,j}}\left(\hat{p}_m\cdot q_{i,j}^m\right)}{\sum_{m\in\mathcal{N}_{i,j}} q_{i,j}^m} \tag{9}$$

where $\hat{f}_{i,j}$ and $\hat{p}_{i,j}$ are, respectively, the reconstructed LGRP feature and the position of pixel $(i, j)$. $\sum_{m\in\mathcal{N}_{i,j}} q_{i,j}^m$ is the normalization constant, meaning that the total probability that a pixel belongs to all superpixels is one. Let $\widehat{\mathbf{Q}}$ indicates the normalize $\mathbf{Q}$ meeting $\sum_{m\in\mathcal{N}_{i,j}} q_{i,j}^m = 1$. Equations (8) and (9) can be rewritten as

$$\hat{\Phi} = \widehat{\mathbf{Q}}\begin{bmatrix} \hat{f}_1 & \hat{p}_1 \\ \vdots & \vdots \\ \hat{f}_M & \hat{p}_M \end{bmatrix} = \widehat{\mathbf{Q}}\tilde{\mathbf{Q}}^{\mathrm{T}}\Phi \tag{10}$$

where $\hat{\Phi} \propto \mathbb{R}^{H\times W\times(R+2)}$ is the reconstructed LGRP and position feature maps of all pixels. The element at the $i$th row, $j$th column of $\hat{\Phi}$ is an $R$ + 2-D vector, representing the feature and location of pixel $(i, j)$. If our superpixel generation network can get an accurate association matrix $\mathbf{Q}$, that is, pixels with very similar features can be merged into the same superpixels, the feature of the $m$th superpixel obtained according to (7) will be very similar to those of the pixels it contains. Furthermore, the reconstructed pixel feature matrix $\hat{\Phi}$ obtained by (10) will be very close to the initial feature matrix $\Phi$. Hence, we define the reconstruction loss in (11) to encourage the superpixel generation model to group pixels with similar property as far as possible.

$$L_{\text{recons}} = \mathcal{L}\left(\hat{\Phi}, \Phi\right) = \mathcal{L}\left(\widehat{\mathbf{Q}}\tilde{\mathbf{Q}}^{\mathrm{T}}\Phi, \Phi\right) \tag{11}$$

where $\mathcal{L}(.,.)$ denotes the distance function. We use the cross-entropy loss function for $\mathcal{L}$ in the task of superpixel generation.

Inspired by SLIC, we also introduce the compactness loss to encourage the superpixels to be spatially compact, which is defined as

$$\begin{aligned} L_{\text{compact}} &= \frac{\sum_{m=1}^{M}\sum_{(i,j)\in\mathbf{I}} \|\hat{p}_m - p_{i,j}\|_2}{H\cdot W\cdot M} \\ &= \frac{\sum_{m\in\mathcal{N}_{i,j}}\sum_{(i,j)\in\mathbf{I}} \|\hat{p}_m - p_{i,j}\|_2}{H\cdot W\cdot|\mathcal{N}_{i,j}|} \end{aligned} \tag{12}$$

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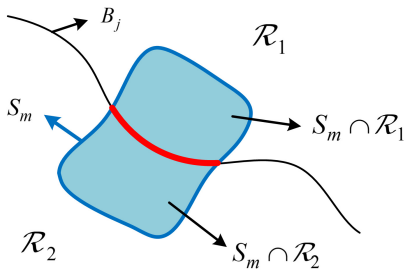



Fig. 2. Illustration of boundary-aware reconstruction loss. $S_m$ refers to the $m$th superpixel (in blue). $\mathcal{R}_1$ and $\mathcal{R}_2$ are regarded as the two classes of landcover regions. $B_j$ is the ground-truth boundary.

where $\|\cdot\|_2$ is the L2-norm. The compactness loss can minimize the spatial distance between the coordinates of superpixel center and all pixels inside it. Hence, the superpixel will have lower spatial variance.

### C. Boundary-Aware Reconstruction Loss

Let $R$ denote the pixel-wise ground truth of a training image. We first convert the ground truth landcover map into the boundary map. As shown in Fig. 2, $\mathcal{R}_1$ and $\mathcal{R}_2$ indicate two adjacent regions with different classes of landcover, respectively. We regard the pixels located at the border of $\mathcal{R}_1$ and $\mathcal{R}_2$ as the boundaries, denoted as $B_j$. In the boundary map, the boundary pixels are ones and the remaining pixels are zeros.

According to the positional relationship between superpixels and boundaries, the superpixels in SAR images can be classified into two categories: the first one is located inside the landcover regions, and the other part is located near or on the boundaries. Whether in segmentation or landcover classification task, the latter type of superpixel obviously should have a greater impact on the final results. For example, in the superpixel-wise water region detection, the accuracy of the water area shape is mainly determined by the superpixels located at the junction of the water area and the land.

The number of two types of superpixels in SAR images is imbalanced, that is, the superpixels far from boundaries are much more than the superpixels on the boundaries. However, the reconstruction error in (11) does not distinguish between these two types of superpixels. This easily causes the superpixel generation model to give priority to improving the accuracy of superpixels inside the landcover, while ignoring the accuracy of superpixels on boundaries. Furthermore, the multiplicative speckle within SAR images may form false-edge pixels in the scenes with low gradient variations. The reconstruction error in (11) may guide the superpixel prediction network to learn false boundary information.

To solve this problem, we propose boundary-aware reconstruction loss function in this section. A boundary-ware weight vector $\{\omega_1, \ldots, \omega_M\}$ is introduced into the reconstruction loss in (11). Specifically, in each training iteration, given the association matrix $\mathbf{Q}$, we can obtain the superpixels by assigning each pixel to the grid cell with the highest probability

$$\eta_{i,j} = \arg\max_m q_{i,j}^m \tag{13}$$

where $\eta_{i,j} \in \{1, \ldots, M\}$ represents the sequence number of the superpixel to which the pixel $(i, j)$ belongs.

Then, we can calculate a boundary-ware weight $\omega_m$ for each superpixel. As illustrated in Fig. 2, $S_m$ refers to the $m$th superpixel, which is located on the boundary $B_j$. The number of pixels that are both in $S_m$ and $\mathcal{R}_1$ ( or $\mathcal{R}_2$) is represented by $|S_m \cap \mathcal{R}_1|$ (or $|S_m \cap \mathcal{R}_2|$). If the predicted $m$th superpixel has no intersection with boundaries, its boundary-aware weight vector $\omega_m$ is set as zero. Otherwise, the boundary-aware weight vector of $S_m$ is defined as

$$\omega_m = \frac{\min\{|S_m \cap \mathcal{R}_1|, |S_m \cap \mathcal{R}_2|\}}{|S_m|}. \tag{14}$$

When $|S_m \cap \mathcal{R}_1|$ is larger than $|S_m \cap \mathcal{R}_2|$, it means superpixel $S_m$ overlaps most with $\mathcal{R}_1$. For an ideal superpixel, its most part will belong to a single region, and $\omega_m$ will approach to zero. On the contrary, an incorrect superpixel will have large intersecting areas with both $\mathcal{R}_1$ and $\mathcal{R}_2$, leading to a large $\omega_m$.

After that, we can map $\omega_m$ into the pixels inside $S_m$, expressed as $\omega_{i,j}^b$, which composes the pixel-wise boundary-aware weight matrix $\mathbf{W}^b \propto \mathbb{R}^{H \times W}$. Finally, the boundary-aware reconstruction loss $L_{\text{recons}}^b$ is defined as

$$L_{\text{recons}}^b = \mathbf{W}^b * \mathcal{L}(\hat{\Phi}, \Phi) = \mathbf{W}^b * \mathcal{L}\left(\widehat{\mathbf{Q}\tilde{\mathbf{Q}}^{\mathrm{T}}}\Phi, \Phi\right) \tag{15}$$

where $*$ represents the Hadamard product of two matrices. If the superpixel is not perfectly adhered with the boundaries, the pixels inside it will have larger $\omega_{i,j}^b$. The boundary-aware reconstruction loss value will be larger, inducing stronger gradients during training with the back-propagation algorithm. Furthermore, since we only weigh the superpixel on ground-truth boundaries, we implicitly handle the data imbalance issue. In the training process, each superpixel will gradually adjust its shape according to the edge information.

### D. Superpixel-Wise Segmentation With Soft Graph Convolution

We aim to design a superpixel-wise end-to-end trainable deep network, which takes the SAR images as input and directly outputs the segmentation results. The superpixel generation network described in the third subsection can over-segment an image into many superpixels. These superpixels form an undirected graph model, where each superpixel corresponds to a vertex and those nodes sharing the common boundaries are connected by edges. This graph model belongs to the non-Euclidean structure data which conventional CNN cannot directly process.

The traditional graph convolution operations can operate on the graph structure data by generalizing convolutions to the graph domain. Each pixel should be assigned uniquely to one superpixel, i.e., $q_{i,j}^m$ is 0 or 1. However, the element $q_{i,j}^m$ of our soft association map is a probability value, denoting the probability of the assigning pixel $(i, j)$ to the $m$th superpixel. In this case, GCN cannot get the required adjacency matrix and degree matrix of the graph model.

We propose a Soft-GCN to solve this problem in this section. First, we need to model the over-segmented SAR image into an undirected graph model $G(V, E)$, where vertices $V$ and edges $E$ represent the superpixels and their position

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relation, respectively. Each vertex in the graph is only connected with its adjacent vertices by edges. The superpixel prediction results show that although the superpixel generation network has adjusted the shapes of superpixels according to the edge information, the relative position of the superpixels basically remains unchanged. That is to say, adjacency matrix $\mathcal{A}$ and degree matrix $\mathcal{D}$ only depend on the initial grid cells. This will avoid recalculating the matrixes $\mathcal{A}$ and $\mathcal{D}$ every time $\mathbf{Q}$ is updated, so greatly reduce the calculation burden of the graph convolution operations.

According to the corollary about the linear formulation of a graph convolution layer given by Kipf *et al.* [42], a superpixel-wise graph convolution operation can be defined as

$$\begin{bmatrix} G_1(\hat{f}_1) \\ \vdots \\ G_1(\hat{f}_M) \end{bmatrix} = \mathcal{G}_{1,\theta} \otimes \begin{bmatrix} \hat{f}_1 \\ \vdots \\ \hat{f}_M \end{bmatrix} = \left(\mathbf{I}_M + \mathcal{D}^{-\frac{1}{2}}\mathcal{A}\mathcal{D}^{-\frac{1}{2}}\right) \begin{bmatrix} \hat{f}_1 \\ \vdots \\ \hat{f}_M \end{bmatrix} \mathcal{G}_{1,\theta}^{\mathrm{T}} \tag{16}$$

where $[\hat{f}_1, \ldots, \hat{f}_M]$ denotes the superpixels' feature set obtained by the superpixel generation network. $G_1(\hat{f}_m) \in \mathbb{R}^{1\times K}$ is the represent of the $m$th superpixel after one time convolution operation. $\mathcal{G}_{1,\theta} \in \mathbb{R}^{K\times R}$ denotes $K$ learnable convolution kernels and $\mathbf{I}_M \in \mathbb{R}^{M\times M}$ is a unit matrix. $\mathbf{I}_M + \mathcal{D}^{-1/2}\mathcal{A}\mathcal{D}^{-1/2}$ is also known as the generalized graph Laplacian. Kipf *et al.* [42] have proved that the computational complexity of this layer-wise graph convolution is $O(|\varepsilon|FC)$, where $\varepsilon$ represents the number of edges. According to (8)–(10), we can further rewrite (16) as follows:

$$\begin{aligned} \begin{bmatrix} G_1(\hat{f}_1) \\ \vdots \\ G_1(\hat{f}_M) \end{bmatrix} &= \left(\mathbf{I}_M + \mathcal{D}^{-\frac{1}{2}}\mathcal{A}\mathcal{D}^{-\frac{1}{2}}\right)\widehat{\mathbf{Q}}\widetilde{\mathbf{Q}}^{\mathrm{T}} \begin{bmatrix} f_{1,1} & \cdots & f_{1,W} \\ \vdots & \ddots & \vdots \\ f_{H,j} & \cdots & f_{H,W} \end{bmatrix} \mathcal{G}_{1,\theta}^{\mathrm{T}} \\ &= \left(\mathbf{I}_M + \mathcal{D}^{-\frac{1}{2}}\mathcal{A}\mathcal{D}^{-\frac{1}{2}}\right)\widehat{\mathbf{Q}}\widetilde{\mathbf{Q}}^{\mathrm{T}}\Phi_{\hat{f}}\mathcal{G}_{1,\theta}^{\mathrm{T}} \end{aligned} \tag{17}$$

where $\Phi_{\hat{f}} \propto \mathbb{R}^{H\times W\times R}$ is a feature matrix, namely

$$\Phi_{\hat{f}} = \begin{bmatrix} f_{1,1} & \cdots & f_{1,W} \\ \vdots & \ddots & \vdots \\ f_{H,j} & \cdots & f_{H,W} \end{bmatrix} \tag{18}$$

where element $f_{i,j}$ denotes the $R$-dimensional LGRP feature vector of pixel $(i, j)$. $G_1(\hat{f}_m)$ is mathematically the weighted sum of its adjacent node features. It means that after the graph convolution operation, the central node will learn the significant task-specific context information from the surrounding nodes. In the follow-up experiments of this article, the GCN we used contains a total of two convolution layers. The output of the second graph convolution operation can be written as

$$\begin{aligned} \begin{bmatrix} G_2(\hat{f}_1) \\ \vdots \\ G_2(\hat{f}_M) \end{bmatrix} &= \left(\mathbf{I}_M + \mathcal{D}^{-\frac{1}{2}}\mathcal{A}\mathcal{D}^{-\frac{1}{2}}\right) \begin{bmatrix} G_1(\hat{f}_1) \\ \vdots \\ G_1(\hat{f}_M) \end{bmatrix} (\mathcal{G}_{2,\theta})^{\mathrm{T}} \\ &= \left(\mathbf{I}_M + \mathcal{D}^{-\frac{1}{2}}\mathcal{A}\mathcal{D}^{-\frac{1}{2}}\right)^2 \widehat{\mathbf{Q}}\widetilde{\mathbf{Q}}^{\mathrm{T}}\Phi_{\hat{f}}(\mathcal{G}_{1,\theta})^{\mathrm{T}}(\mathcal{G}_{2,\theta})^{\mathrm{T}} \end{aligned} \tag{19}$$

where $\mathcal{G}_{2,\theta} \in \mathbb{R}^{C\times K}$ is the learnable parameter matrix of the second graph convolution layer. $C$ indicates the number of convolution kernels, which is also equal to the number of land-covers. With the layers being deeper, the input LGRP features are gradually aggregated to high-level abstract representations. Finally, we use the SoftMax activation function to normalize $G_2(\hat{f}_m)$. Therefore, $G_2(\hat{f}_m)$ is a $1 \times K$ dimensional vector and its $\upsilon$th element means the probability that node $m$ belongs to class $\upsilon$.

We use a majority voting strategy to obtain the labels of superpixels. $R$ denotes the pixel-wise ground truth of a training image. The most frequent class in the $m$th superpixel will be selected as the category of this superpixel. $y_m$ indicates the one-hot encoding labels of the $m$th superpixel and $\mathbf{Y} = [y_1, \ldots, y_M]$ denotes the label matrix. The GCN's loss function is defined as

$$\begin{aligned} &L_{\text{graph}} \\ &= \mathcal{L}\left\{\begin{bmatrix} G_2(\hat{f}_1) & \cdots & G_2(\hat{f}_M) \end{bmatrix}^{\mathrm{T}}, [y_1, \ldots, y_M]^{\mathrm{T}}\right\} \\ &= \mathcal{L}\left\{\left(\mathbf{I}_M + \mathcal{D}^{-\frac{1}{2}}\mathcal{A}\mathcal{D}^{-\frac{1}{2}}\right)^2 \widehat{\mathbf{Q}}\widetilde{\mathbf{Q}}^{\mathrm{T}}\Phi_{\hat{f}}(\mathcal{G}_{1,\theta})^{\mathrm{T}}(\mathcal{G}_{2,\theta})^{\mathrm{T}}, \mathbf{Y}^{\mathrm{T}}\right\}. \end{aligned} \tag{20}$$

Note that $\mathcal{D}$, $\mathcal{A}$, $\Phi_{\hat{f}}$, and $y_m$ in (20) are decided by the input SAR images and the ground truth, whereas $\widehat{\mathbf{Q}}$, $\widetilde{\mathbf{Q}}$, $\mathcal{G}_{1,\theta}$, and $\mathcal{G}_{2,\theta}$ are derived from the parameters of superpixel generation and GCNs.

To sum up, we can give the overall loss function of the proposed superpixel-wise segmentation network

$$\begin{aligned} &L_{\text{class}} \\ &= \left(L_{\text{recons}} + L_{\text{recons}}^b\right) + \alpha \cdot L_{\text{compact}} + \beta \cdot L_{\text{graph}} \\ &= \left(1+\mathbf{W}^b\right)\mathcal{L}\left(\widehat{\mathbf{Q}}\widetilde{\mathbf{Q}}^{\mathrm{T}}\Phi, \Phi\right) + \alpha \cdot \frac{\sum_{m\in\mathcal{N}_{i,j}}\sum_{(i,j)\in\mathbf{I}} \|\hat{p}_m - p_{i,j}\|_2}{H \cdot W \cdot |\mathcal{N}_{i,j}|} \\ &\quad + \beta \cdot \mathcal{L}\left\{\left(\mathbf{I}_M + \mathcal{D}^{-\frac{1}{2}}\mathcal{A}\mathcal{D}^{-\frac{1}{2}}\right)^2 \widehat{\mathbf{Q}}\widetilde{\mathbf{Q}}^{\mathrm{T}}\Phi_{\hat{f}}(\mathcal{G}_{1,\theta})^{\mathrm{T}}(\mathcal{G}_{2,\theta})^{\mathrm{T}}, \mathbf{Y}^{\mathrm{T}}\right\} \end{aligned} \tag{21}$$

where $\alpha$ and $\beta$ are the weights of compactness loss and graph convolution loss to balance the three terms, respectively. In our work, $L_{\text{recons}}$ and $L_{\text{recons}}^b$ use the $\ell_2$-norm as the distance measure. Similar to [38], we adopt cross-entropy to calculate $L_{\text{graph}}$ to penalize the difference between the network output and the labels of the original labeled examples. It can be seen from (21) that $L_{\text{recons}}$, $L_{\text{recons}}^b$ and $L_{\text{compact}}$ aim at adjusting the superpixels' shape according to the boundary information in the SAR image. What is more, $L_{\text{graph}}$ can update the parameters of the graph convolution kernel to improve the accuracy of the segmentation results on one hand. On the other hand, it can adjust the parameters of the superpixel generation network according to the segmentation results. Fig. 3 and Table I give the flowchart and implementation details of our proposed superpixel-wise segmentation algorithm.

## IV. Experimental Results and Analysis

To verify the proposed approach, we conduct the superpixel generation and segmentation experiments on both the simulated and real SAR images in this section. To be specific,

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TABLE I
TRAINING STEPS OF DSSG

**Input:** SAR image $\mathbf{I} \propto \mathbb{R}^{H \times W}$
**Output:** Superpixel-wise segmentation results

Extract $R$-dimensional LGRP feature vector of pixel $(i,j)$, i.e., $\mathrm{f}_{i,j}$.
**for** Number of training iterations **do**

1. The superpixel generation network takes *LGRP-position* feature maps $\Phi$ as input.
2. The superpixel generation network outputs soft association maps $\mathbf{Q}$.
3. $\mathbf{Q}$ is the normalized to make, denoted as $\tilde{\mathbf{Q}}$.
4. Update the superpixel centers as (7), denoted as $\hat{\mathrm{f}}_m$, $\hat{\mathrm{p}}_m$.
5. Reconstruct the *LGRP-position* features of all pixels as (8) (9), represented as $\hat{\Phi}$.
6. Calculate reconstruction loss ($L_{recons} + L^b_{recons}$) as (11), (15) and compactness loss $L_{compact}$ as (12).
7. Model the over-segmented SAR image into an undirected graph model $G(V,E)$.
8. Graph convolution network outputs the segmentation results $\mathrm{G}_2(\hat{\mathrm{f}}_m)$.
9. Calculate segmentation loss function as (20), defined as $L_{graph}$.
10. Update the parameters of networks based on the $L_{recons}$, $L^b_{recons}$, $L_{compact}$ and $L_{graph}$.

**End for**

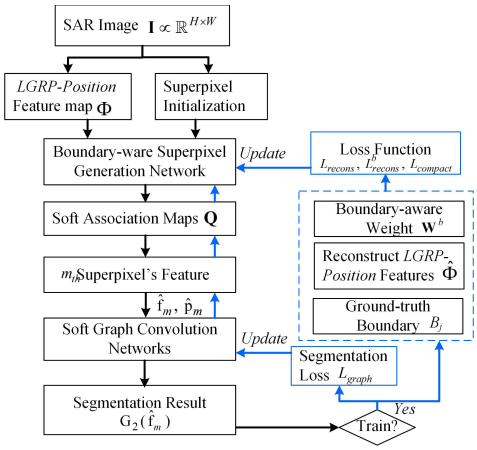



Fig. 3. Flowchart of the proposed algorithm. The black arrow indicates forward propagation, and the blue arrow indicates backward propagation.

we first introduce the used datasets and parameter settings. Then, we compare our superpixel generation and segmentation results with other existing methods. Finally, we give the running time of different methods on the simulated and real datasets.

### A. Dataset and Evaluation Metrics

Three real SAR databases are used to demonstrate the segmentation performance of the proposed method, i.e., Fangchenggang, Poyang Lake, and Gaofen dataset.

*1) Fangchenggang:* The Fangchenggang dataset is acquired by airborne SAR sensors (Ku-band, HV polarization) at Fangchenggang Town (N21°41′22.66″, E108°21′2.64″), Guangxi, China, in March 2016. The image size is 8515 × 6736 pixels and the resolution is 2 m. This dataset consists of three interested landcover classes: farmland, water, and urban. The remaining regions are regarded as background.

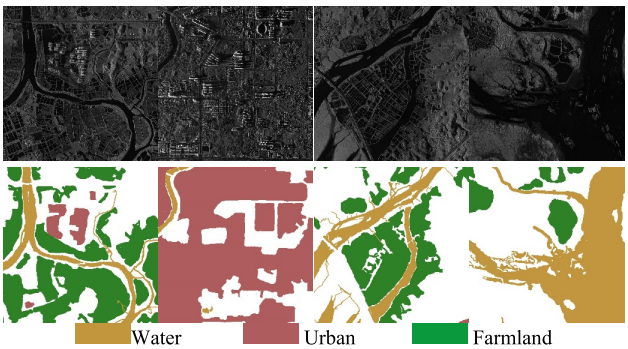



Fig. 4. Four exemplar images of the Fangchenggang dataset and the corresponding ground-truth maps.

TABLE II
NUMBERS OF TRAINING AND TESTING PIXELS OF FOUR CLASSES IN FANGCHENGGANG DATASET

| Class | Farmland | Water | Urban | Background |
|---|---|---|---|---|
| Train | 1,734,354 | 3,165,411 | 1,846,234 | 23,348,065 |
| Test | 1,946,283 | 3,534,694 | 1,738,216 | 20,043,783 |

Fig. 4 shows four exemplar SAR images and their ground truth. The yellow, red, green, and white regions represent water, urban, farmland, and background regions, respectively. Note that the SAR scene is cropped into 1024 × 1024 images. Table II gives the proportions of the four classes of pixels in the training set and testing set.

*2) Poyang Lake:* The Poyang Lake dataset contains two SAR images, which is taken by spaceborne GF-3 sensors operating in the L-band, FSII mode, and HH single polarization. The imaging range is at Poyang Lake (E116.2, N29.2), Jiangxi, China, in July 2020, which is shown in Fig. 5. Two images' spatial resolution is 10 m × 10 m, and their sizes are 28 744 × 26 675 and 27 571 × 29 823, respectively. As largest freshwater lake in China, Poyang Lake has complex geometric shapes, which requires the superpixels to accurately capture the boundaries of water regions. This dataset is labeled manually according to Google Earth maps, including two landcover classes: "Water" and "Land." Similarly, two SAR scenes are cropped into 1024 × 1024 images, where 30% of them are randomly selected as the trainset and the others belong to the test set.

*3) Gaofen Dataset:* The Gaofen dataset was released in 2020 Challenge on Automated High-Resolution Earth Observation Image Interpretation [43]. It contains 500 high-resolution spaceborne SAR images with L-band, FSII mode, and HV single polarization. The size of each image is 512 × 512, and the resolution range is 10–30 m. The ground-truth maps are annotated with the six recognized landcover types: water body, buildings, industrial area, lawn, bare land, and background. We randomly select 100 images as training samples and the remaining are used as test samples.

*4) Evaluation Metrics:* To quantitatively compare the performance of various methods, we introduce four indicators: F1-score [44], Cohen's kappa coefficient ($\kappa$) [44], pixel-wise Overall Precision (OP) [45], and pixel-wise Overall Accuracy (OA) [46]. Specifically, Recall is the ratio between the pixels correctly predicted in a certain class against the total number of pixels of this class. Precision corresponds to the proportion

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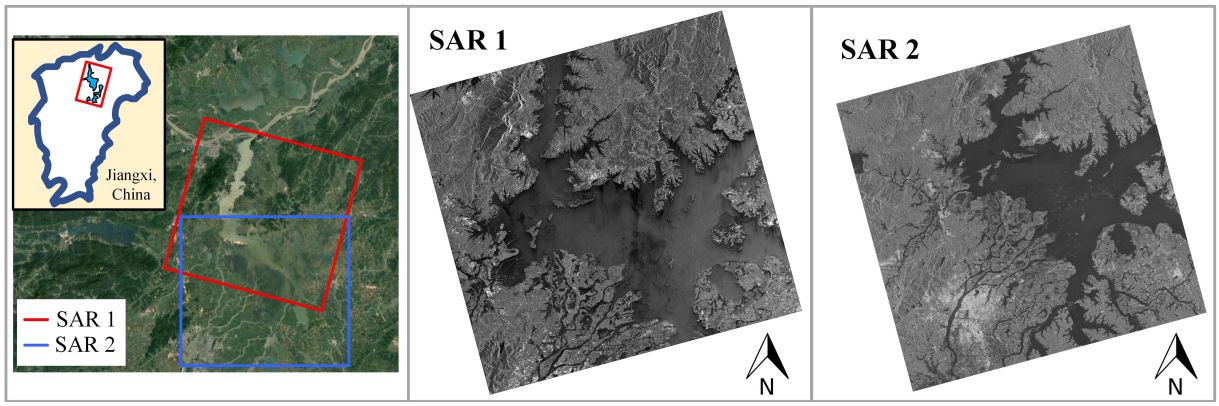



Fig. 5. Location of the Poyang Lake dataset. (a) Observed area of two SAR images in the Poyang Lake dataset. (b) First SAR image with the size of 28 744 × 26 675. (c) Second SAR image with the size of 27 571 × 29 823.

of the pixels segmented in a certain class that are actually right. OP and OA are the weighted average of recall and precision, respectively, where the weights indicate the proportion of each class of pixels in the ground-truth images. The F1-score is defined based on precision and recall as

$$F_1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{22}$$

The closer the segmentation result is to the ground truth, the greater the value of F1-score. The kappa coefficient is defined as

$$\text{Kappa} = \frac{p_o - p_e}{1 - p_e} \tag{23}$$

where $p_o$ is the relative agreement between the segmentation results and the real labels; $p_e$ is the hypothetical probability of a chance agreement. A kappa coefficient closer to one denotes better agreement between the results and ground truth.

*5) Implementation Details:* Our superpixel generation network uses an encoder–decoder design with skip connections, which is the same with [38], consisting of five convolution layers and four deconvolution layers. The five convolution layers contain 16, 32, 64, 128, and 256 convolution kernels with a size of 3 × 3, respectively. We use a grid with cell size 16 × 16 as the initial superpixels. The Soft-GCN consists of two graph convolution layers, which is identical to [24]. The first spectral convolution layer has eight convolution filters and the second layer outputs the classification probability vector, so the number of filters remains consistent with that of the landcover types. Two sub-networks take Leaky ReLU as activation function except their output layers, which use Softmax. The model is implemented with PyTorch and optimized using Adam with learning rate $= 1 \times 10^{-4}$ and batch size $= 100$. Convergence is reached at about 200 k iterations. According to the suggestions in [23] and [38], we set the weights of compactness loss and graph convolution loss as $\alpha = 0.001$ and $\beta = 0.1$, respectively, in all experiments.

### *B. Superpixel Generation Results for the Synthetic and Real SAR Datasets*

This part will verify the effectiveness of the proposed deep superpixel generation network on synthetic images.

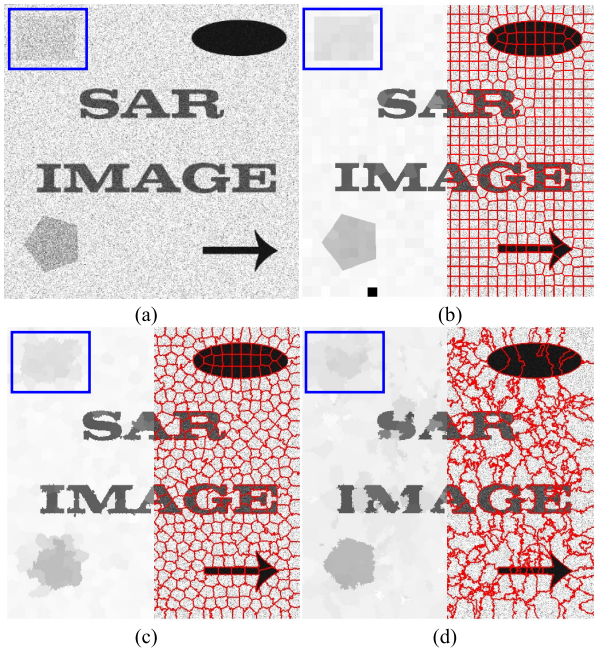



Fig. 6. Simulated SAR image with shape texture and superpixel generation results. (a) Simulated SAR image with four-look Gamma speckle. (b)–(d) Superpixel generation results by our method and PILS and SLIC-GGD methods. The red lines in the right part denote the superpixels, and the pixel value in the left part is the mean value of the superpixel to which it belongs.

The simulated SAR images are contaminated by multiplicative speckle with different numbers of looks. Specifically, we first design the noise-free images with different shape texture and carton texture. Then, we add the multiplicative speckle obeying Gamma distributed to the designed images to simulate the real SAR images. Note that we set the equivalent number of looks of the speckle to four.

Figs. 6 and 7, we use the proposed method to over-segment the simulated images and compare them qualitatively with other methods. Figs. 6(a) and 7(a) shows two simulated SAR images with a size of 512 × 479. In Figs. 6(b) and 7(b), the right part denotes the superpixel boundary and the pixel value in the left part is the mean value of the superpixel to which it belongs. The results of two superpixel generation algorithms, namely, pixel intensity and location similarity (PILS) [21] and SLIC with generalized gamma distribution (SLIC-GGD) [47],

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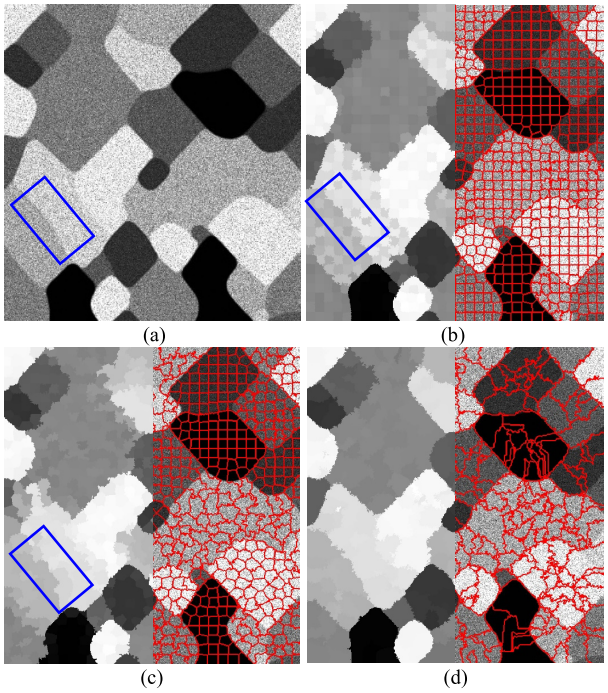



Fig. 7. Simulated SAR image with carton texture and superpixel generation results. (a) Simulated SAR image with four-look Gamma speckle. (b)–(d) Superpixel generation results by our method and PILS, and SLIC-GGD method. The red lines in the right part denote the superpixels, and the pixel value in the left part is the mean value of the superpixel to which it belongs.

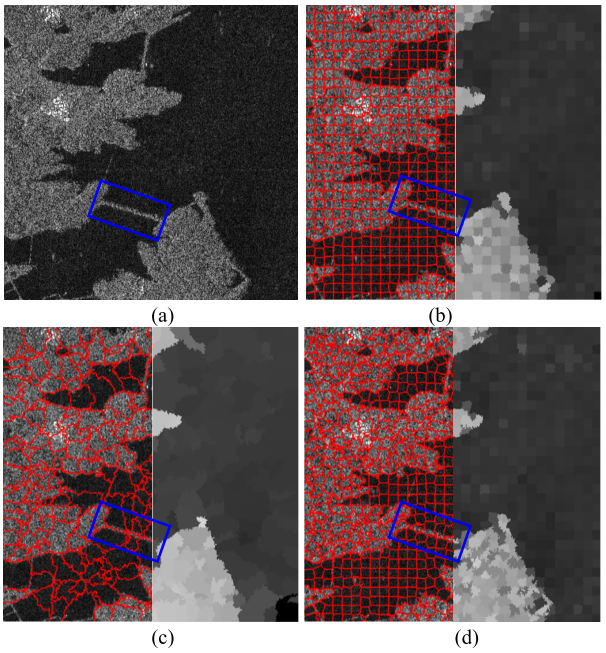



Fig. 8. Real SAR image and superpixel generation results. (a) Real SAR image. (b)–(d) Superpixel generation results by our method and PILS and SLIC-GGD methods. The red lines in the left part denote the superpixels, and the pixel value in the right part is the mean value of the superpixel to which it belongs.

are given in Figs. 6 and 7 as compared method. PILS is an SAR image over-segmentation method using the PILS to cluster the neighbor and similar pixels into one superpixel. The parameters of PILS and SLIC-GGD are consistent with those suggested in the corresponding articles [21] and [47], respectively.

The gray rectangular in the upper left corner of Fig. 6(a) is the most difficult area to segment, which has been marked by a blue rectangular. The contrast between this rectangular area and the background is inherently small, and the multiplicative noise makes the borders of the rectangle blurred. In Fig. 6(b), our method can accurately retain the contour information of this rectangular area, whereas the superpixels of PILS and SLIC-GGD cannot well-adhere the real boundaries of rectangular.

Similarly, we mark the boundary that is difficult to extract using blue rectangles in Fig. 7. Our method makes full use of the feature extraction and integration capabilities of the deep convolution network to accurately extract the boundaries. However, for PILS and SLIC-GGD, the superpixels based on the hand-crafted features cannot retain the boundary information well.

This section also gives the superpixel generation results of different methods for real SAR image in Fig. 8. There is a $1024 \times 1024$ image from the Poyang Lake dataset in Fig. 8(a). Note that this scene includes winding coastlines and bridges, which is a daunting challenge for superpixel estimation. From Fig. 8(b), it can be observed that our superpixels are more compact in homogeneous areas and are shape-adaptive in non-homogeneous scenes, thanks to the adaptive feature extraction and fusion capability of deep convolution network. Since the superpixel generation network can learn the edge details and image structure from the ground truth of the segmentation task, it can well-preserve the shape of bridge in the blue rectangular. In contrast, from Fig. 8(c) and (d), we can find that the PILS and SLIC-GGD's superpixels have lower boundary adherence. They cannot detect the boundary of bridge and generate some regions across various landcovers. These comparative experiments above all prove that the superpixels generated by our method can better retain the boundary information in the SAR images, which will improve the accuracy of the subsequent SAR segmentation.

To quantitatively evaluate the results of various superpixel generation methods, here we introduce boundary recall (BR). BR is defined as the ratio of ground-truth boundaries which are recovered by the superpixel edges correctly. A higher BR value means that superpixels can better adhere the true boundary and less true boundaries are omitted. As shown in Fig. 9, we calculate the BR indicators of PILS, SLIC-GGD, and our DSSG method on the simulated and real SAR datasets. "Simulated 1" and "Simulated 2" in Fig. 9 represent the simulated images in Figs. 6(a) and 7(a), respectively. Note that the BR values of Fangchenggang, Poyang Lake, and Gaofen refer to the average BR of the multiple images in the corresponding dataset. Although the five datasets are quite different, the BR value of our DSSG method can be close to or higher than 0.9, which is better than the other two methods. For example, when over-segmenting the simulated SAR image with shape texture in Fig. 6(a), DSSG's BR value is 0.96, which is better than 0.92 of PILS and 0.86 of SLIC-GGD. In addition, we can find that the BR values of the three methods with real images are lower than the BR values with the simulated images. This is because the scene in the real

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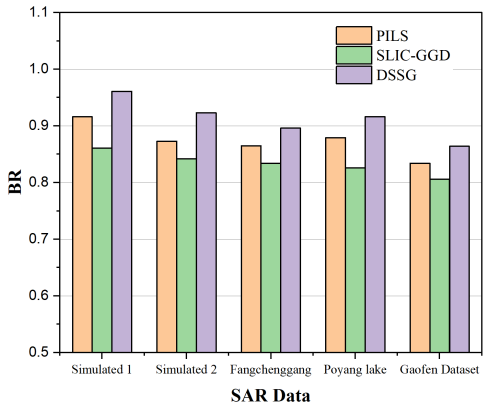



Fig. 9. BR of three superpixel generation methods for different SAR images.

image contains more types of landcovers, more speckle noise, and more complex geometric distortions.

### C. Segmentation Experiments on Fangchenggang Dataset

This section will compare the results of the proposed DSSG with four state-of-the-art methods on the Fangchenggang dataset, i.e., SegNet, Deeplap V3+, PILS and GCN (PILSG), and SLIC-GGD and GCN (SLIC-GGDG). SegNet [48] and Deeplap V3+ [49] are two representative pixel-wise encoder–decoder structure semantic segmentation methods. SegNet's encoder is in charge of feature extraction, composed of 13 convolution layers of VGG16 trained on ImageNet. The decoder sub-network transforms the features to the segmentation maps by multiple upsampling operations. The key parameters of SegNet are set as follows: patch size $= 16 \times 16$, learning rate $= 0.0001$, regular regularization parameter $= 0.0005$, batch size $= 300$, and training epoch $= 100$, which are all consistent with [48]. Deeplap V3+ is a recently proposed state-of-the-art semantic segmentation network, which is mainly composed of ResNet101-V2 and the atrous spatial pyramid pooling (ASPP) module. ResNet101 network with 101 layers has better segmentation performance than the shallow residual network. The ASPP module derivers the concept of pyramid pooling and atrous convolution. It can resist the limitation of the fixed size of the input image to extract high-level multiscale information. Atrous convolution also make the encoder and decoder models more efficient. The hyperparameters of Deeplap V3+ are consistent with [49].

For the PILSG method, PILS is first used to over-segment the SAR image into a set of superpixels. Then we introduce the CNN proposed in [24] to extract the features of superpixels. Finally, GCN is used to predict the categories of superpixels. Similarly, SLIC-GGDG refers to the first use of SLIC-GGD to generate superpixels, which is followed by the GCN-based segmentation. Note that the structure and hyperparameters of the graph convolutional network here are consistent with the proposed method in this article. For PILSG and SLIC-GGDG, the number of superpixels is a significant parameter. Smaller superpixels mean larger number of superpixels, resulting in a decrease in segmentation speed, whereas too large superpixels cannot retain sufficient edge information. Multiple experiments show that two comparison methods can achieve relatively better results when the number of superpixel is set to 4000. In this case, each superpixel has 260 pixels on average.

Table III depicts the F1-score and kappa values of multiple methods for the Fangchengang dataset. DSSG has highest F1-score and kappa values. Taking F1-score as an example, DSSG is 90.33, higher than SegNet (85.38), Deeplab V3+ (83.82), PILSG (89.48), and SLIC-GGDG (89.79). These results demonstrate that our DSSG can capture satisfactory performance for SAR images.

To compare the segmentation results intuitively, Fig. 10 displays the maps of various methods. Fig. 10(a) and (b) shows an input SAR image of the Fangchenggang dataset and the ground truth. We mark some narrow tributaries using the red rectangles in the scene, which are easily ignored by the segmentation algorithm. In addition, due to the side-looking imaging characteristic of the SAR, the mountainous region forms a lot of shadow areas in the image. There is no scattered echo in these shadow areas. Considering the backscattered echo in the river area is also very weak, these two categories of areas are almost indistinguishable without considering the context information, increasing the difficulty of subsequent segmentation. Fig. 10(c) and (d) shows the segmentation result of SegNet and Deeplab V3+. As SegNet predicts each $16 \times 16$ patch separately, its results have much isolated noise and have less spatial consistency. Deeplab V3+ has more parameters than other methods and requires sufficient labeled data. Due to the limited number of training samples in the Fangchenggang dataset, its segmentation results have a lot of errors. In the results of PILSG [Fig. 10(e)] and SLIC-GGDG [Fig. 10(f)], some of the thin rivers are missing, and others' shapes have been changed. On the contrary, our method [Fig. 10(g)] can effectively preserve the contours of these river regions and has fewer areas of confusion. That is because the proposed DSSG can adjust the superpixels' shapes according to both segmentation task and boundary information in input SAR image, which significantly improves the superpixels' quality and segmentation accuracy. Furthermore, the graph convolution can consider more context information than CNN around the central superpixel when segmenting the superpixel set, so it results in better spatial compactness. By comparing the results of SegNet and DSSG, it can be shown that the segmentation accuracy of the superpixel-wise method can be higher than that of the pixel-wise method.

### D. Segmentation Performance on Poyang Lake Dataset

This section aims at verifying the effectiveness of DSSG for images of different SAR sensors. The Poyang Lake SAR images have stronger speckle noise, and the shape of the water area is more irregular than Fangchenggang.

As shown in Table IV, we first quantitatively compare our algorithm with four state-of-the-art methods on the Poyang Lake dataset. The DSSG algorithm obviously achieves the highest F1-score and kappa of 93.24% and 0.861%, respectively, indicating that the edge detection result is satisfactory. SLIC-GGDG essentially is a local clustering scheme combining intensity similarity with spatial proximity, which is easily

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TABLE III
COMPARISON OF THE OVERALL SEGMENTATION ON FANGCHENGGANG DATESET

| Class | SegNet | Deeplab V3+ | PILSG | SLIC-GGDG | DSSG |
|---|---|---|---|---|---|
| Farmland | 83.22±0.12 | 72.85±0.21 | 81.15±0.09 | 81.37±0.11 | 82.91±0.12 |
| Water | 66.79±0.12 | 76.84±0.22 | 61.60±0.08 | 82.28±0.25 | 84.60±0.07 |
| Urban | 66.32±0.48 | 76.84±0.17 | 70.66±0.98 | 82.86±0.23 | 80.02±0.23 |
| OP (%) | 85.85±0.32 | 85.58±0.42 | 89.53±0.54 | 89.94±0.41 | **90.43±0.14** |
| OA (%) | 84.90±0.22 | 82.13±0.26 | 89.43±0.32 | 89.68±0.33 | **90.24±0.20** |
| F1-score (%) | 85.38±0.13 | 83.82±0.22 | 89.48±0.41 | 89.79±0.18 | **90.33±0.05** |
| Kappa | 0.758±0.009 | 0.724±0.005 | 0.827±0.006 | 0.830±0.004 | **0.839±0.003** |

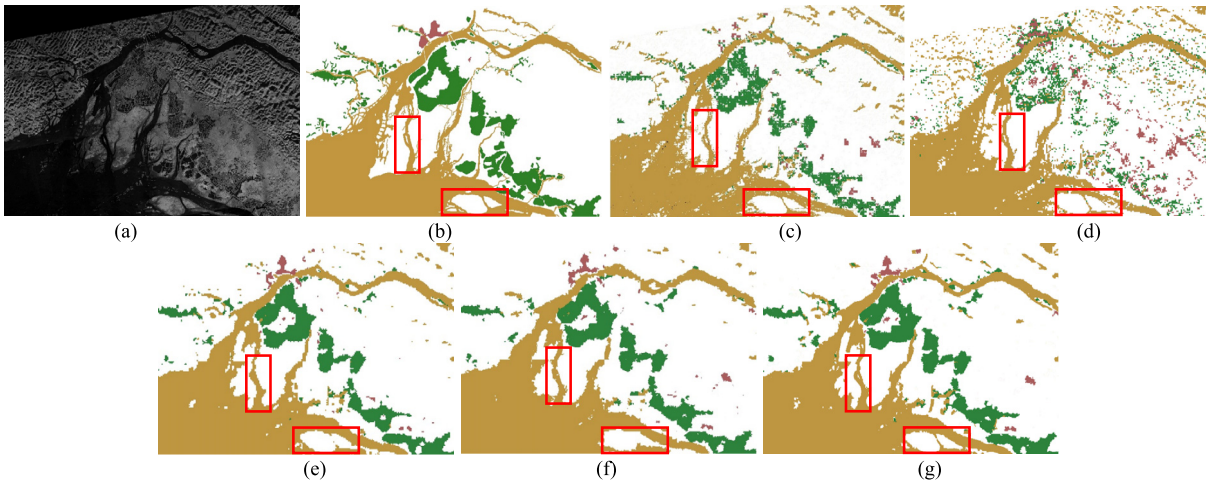



Fig. 10. Some detection results of the proposed method for Fangchenggang dataset. (a) Original SAR images. (b) Ground-truth. (c) SegNet. (d) Deeplab V3+. (e) PILSG. (f) SLIC-GGDG. (g) DSSG.

affected by speckle, and hence it has the minimum F1-score and kappa.

Fig. 11 gives the segmentation results of different methods for a Poyang Lake SAR image, i.e., SegNet, Deeplab V3+, PILSG, and SLIC-GGDG. The parameters of these methods are consistent with the relevant experiments in the Fangchenggang database. Specially, Fig. 11(a) and (b) gives an input SAR image and its ground-truth map. In Fig. 11(c)–(g), four methods for comparison all can obtain obvious edges. However, for the river marked with red rectangles, our method can accurately get the weak and tiny boundaries, so that it retains the shape of river delta. The segmentation result is more similar to the ground-truth map.

Fig. 12 denotes the results of multiple methods on another SAR image in the Poyang Lake dataset. It can be seen that there are a large number of fishponds by the lake, such as the area marked by a red rectangle. These fishponds have similar backscatter coefficients to the lake; hence, they look very similar in the image. There are only weak boundaries between these fishponds and the lake areas, which is a big challenge for the segmentation algorithm. It can be observed that four reference methods can detect most of the distinct boundaries in images. PILSG and SLIC-GGDG mainly use the image intensity texture features for segmentation. Therefore, their results are more easily disturbed by the speckle noise in SAR images. Some weak and thin edges of fishponds are mislabeled as water areas in the marked fishponds areas. As SegNet fails to consider the global context information during convolution operation, there are many isolated mis-segmentation points in SegNet's results. The segmentation performance of Deeplab V3+ on small-scale datasets is not satisfactory. However, our DSSG method in Fig. 12(g) can obtain most of the edges accurately and has fewer confusion. The reason is that our method can modify the shape of superpixels according to the task. Furthermore, the superpixel generation network and the graph-based segmentation network can be trained simultaneously, so their parameters can capture the global optimal values.

### *E. Segmentation Performance on Gaofen Dataset*

Compared with the other two real datasets, it has more classes of landcovers and lower resolution, so it is the more difficult to segment. Fig. 13 shows the segmentation results of some Gaofen images by multiple methods, and Table V gives the corresponding evaluation matrix. Fig. 13(a) and (b) are the input images and the corresponding ground truth. Fig. 13(c) and (d) are the results of SegNet and Deeplab V3+. Among them, SegNet's encoder network has fewer layers and cannot fully extract semantic features, so it misclassifies many water areas as grass and bare land. In Table III, the kappa coefficient of SegNet is also the lowest (0.692). The kappa coefficients of Deeplab V3+, PILSG, and SLIC-GGDG are similar, but much lower than the proposed DSSG. It is worth pointing out that although PILSG and SLIC-GGDG, belonging to the local clustering algorithms, are essentially improvements of conventional SLIC algorithms, they can only take the local context information into consideration and their superpixel generation processes usually fail to

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TABLE IV
COMPARISON OF THE OVERALL SEGMENTATION PERFORMANCE ON POYANG LAKE DATESET

| Class | SegNet | Deeplab V3+ | PILSG | SLIC-GGDG | DSSG |
|---|---|---|---|---|---|
| Water | 92.08±0.01 | 92.68±0.17 | 90.50±0.67 | 90.88±0.41 | **92.24**±0.11 |
| Land | 94.51±0.23 | 93.55±0.24 | 95.44±0.26 | 91.31±0.29 | **95.03**±0.31 |
| OP (%) | 93.78±0.44 | 93.95±0.24 | 93.96±0.34 | 91.18±0.28 | **94.19**±0.21 |
| OA (%) | 93.82±0.22 | 94.69±0.28 | 93.98±0.14 | 91.20±0.43 | **94.23**±0.27 |
| F1-score (%) | 92.94±0.71 | 92.30±0.13 | 92.21±0.46 | 91.04±0.43 | **93.24**±0.10 |
| Kappa | 0.851±0.005 | 0.846±0.008 | 0.856±0.003 | 0.782±0.022 | **0.861**±0.003 |

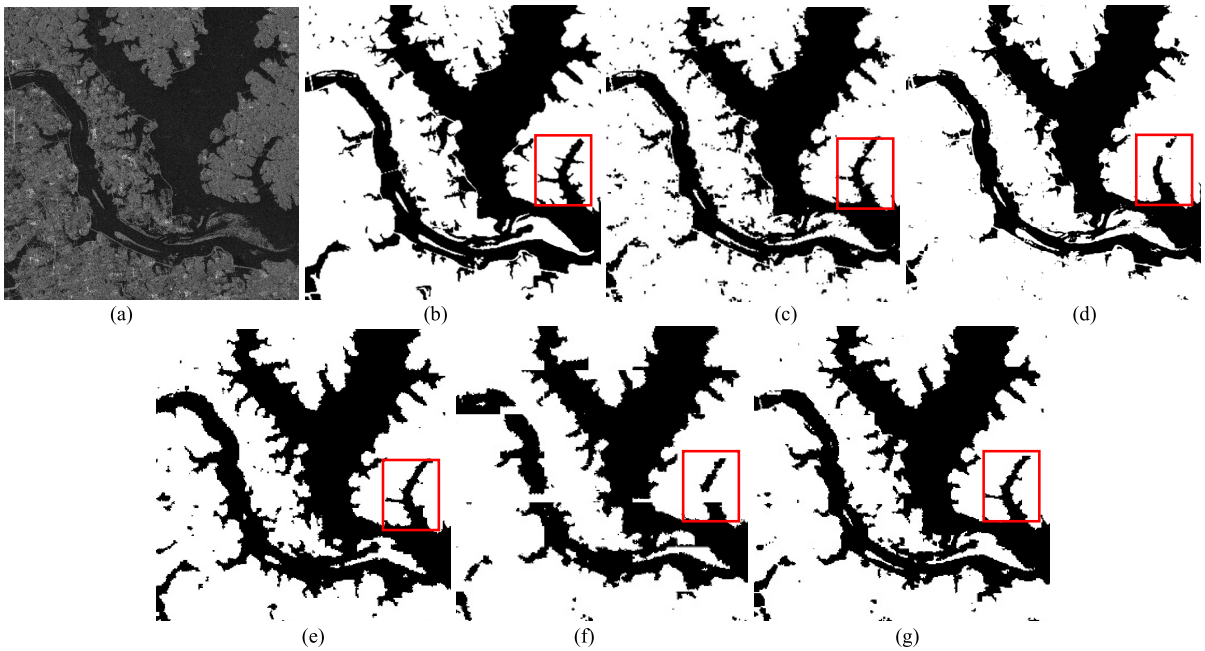



Fig. 11. Some detection results of the proposed method for Fangchenggang dataset. (a) Original SAR images. (b) Ground truth. (c) SegNet. (d) Deeplab V3+. (e) PILSG. (f) SLIC-GGDG. (g) DSSG.

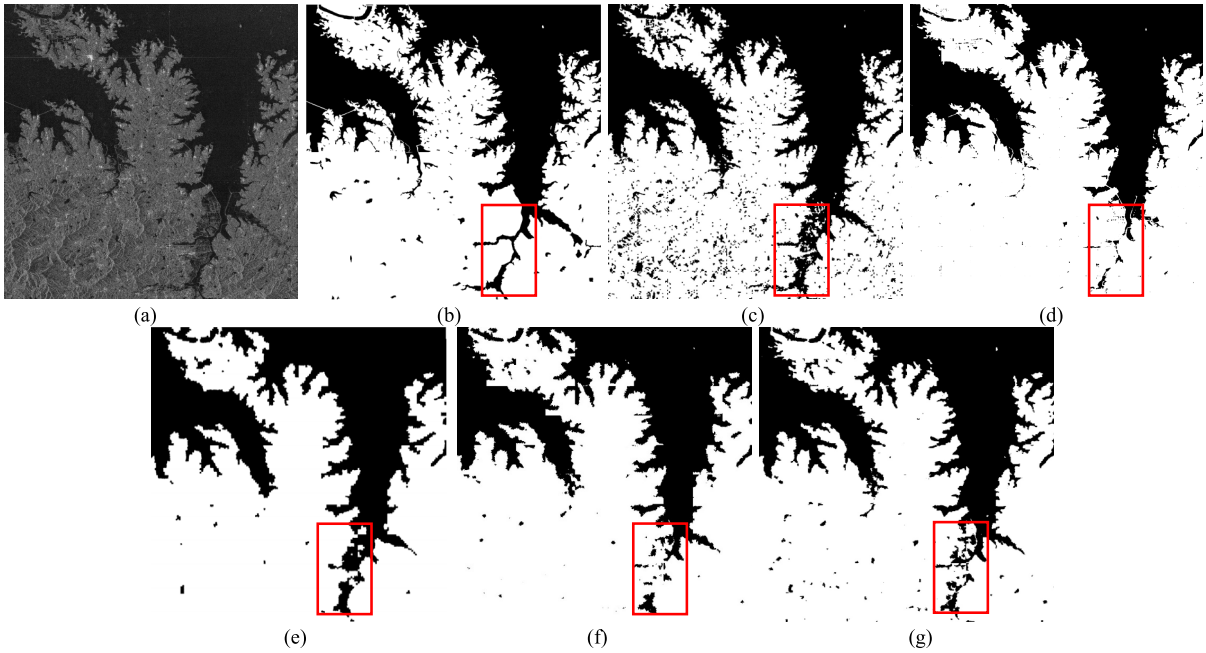



Fig. 12. Some detection results of the proposed method for Poyang Lake dataset. (a) Original SAR images. (b) Ground-truth. (c) SegNet. (d) Deeplab V3+. (e) PILSG. (f) SLIC-GGDG. (g) DSSG.

adhere the weak boundaries. In contrast, the proposed method can learn the global information from the ground truth, so the superpixel can effectively fit the boundaries and correctly segment the different landcovers. These results in Fig. 13(g)

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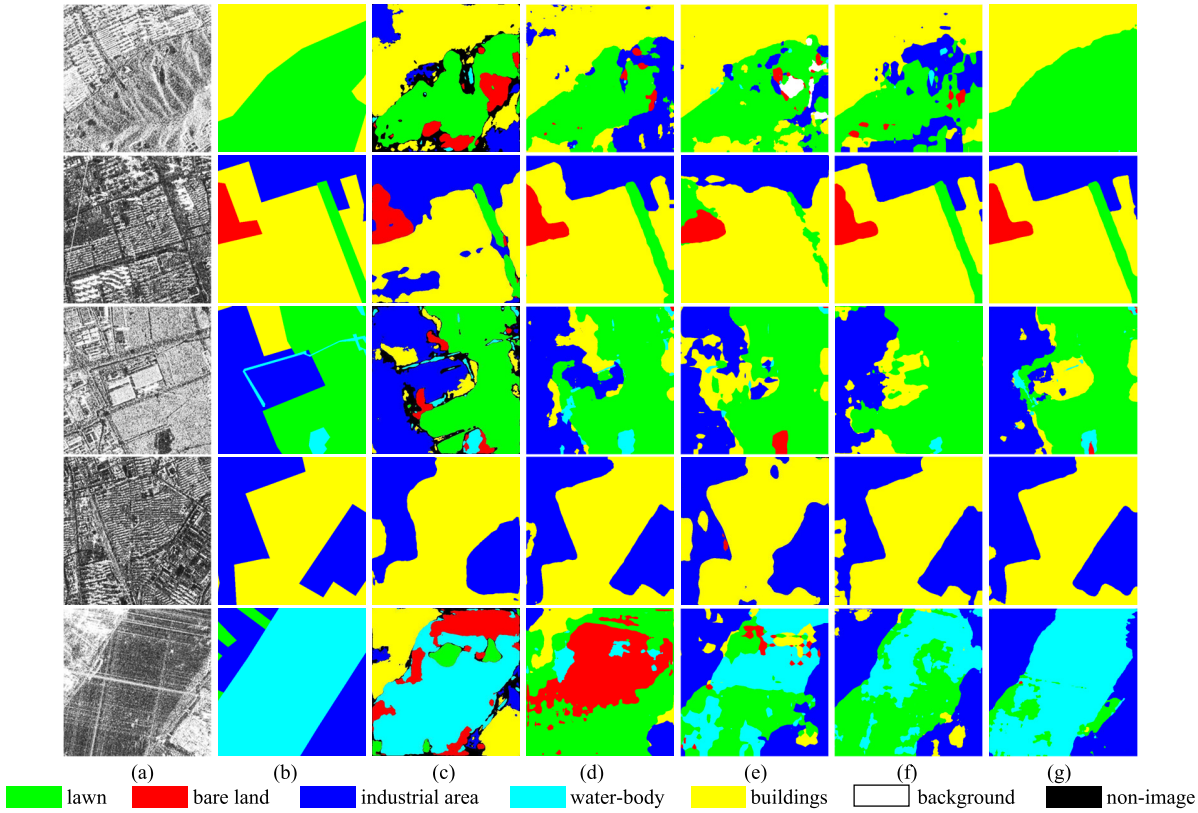



Fig. 13. Some detection results of the proposed method for Gaofen dataset. (a) Original SAR images. (b) Ground-truth. (c) SegNet. (d) Deeplab V3+. (e) PILSG. (f) SLIC-GGDG. (g) DSSG.

demonstrate that the proposed DSSG is more suitable for the SAR image with complex scenes and weak edges.

By comparing the results of Tables III–V, it can be found that the segmentation performance of the same method is not stable when processing different datasets. For example, SegNet's average F1-score is higher than PILSG and SLIC-GGDG on the Poyang Lake data. However, its results are the worst when processing the Fangchenggang data, far lower than the other four methods. The SLIC-GGDG method achieves satisfactory segmentation results for the Fangchenggang data, but its processing results on the Poyang Lake data are indeed the worst among all the methods. On the contrary, our method keeps the best results for three real datasets, showing that it can be applied to various SAR sensors.

### *F. Effect of Initial Superpixel Size on Segmentation Results*

For all superpixel-wise segmentation methods, the initial size of the superpixel is one of the most critical parameters, and it determines the segmentation accuracy to a certain extent. So, in this part we compare the results of multiple methods under different superpixel sizes. Figs. 14–16 depict the kappa and F1-score curves of three methods (PILSG, SLIC-GGDG, and DSSG) with different sizes of superpixels for the three datasets, respectively. Note that the kappa and F1-score values here are the average of the results of all test images. Since SegNet and Deeplab V3+ do not take superpixels as segmentation units, it is not included in Figs. 14–16. The abscissa represents the number of superpixels for each image. It can be seen that the size of the superpixel has a great

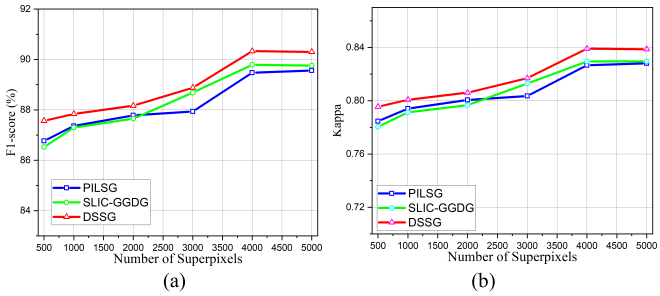



Fig. 14. F1-score and kappa curves with different superpixel numbers for the Fangchenggang dataset. (a) F1-score curves for the Fangchenggang dataset. (b) Kappa curves for the Fangchenggang dataset.

impact on the segmentation results of all the methods. Take Fig. 14 for example, when the number of superpixels is 500, each superpixel contains about 2000 pixels. Too many pixels are assigned to one superpixel so that many weak edges will be easily ignored during superpixel generation, leading to some segmentation errors. As the number of superpixels increases from 500 to 4000, the superpixels gradually become smaller, and the shape of the superpixels can better adhere the edges. So the F1-scores and kappa coefficients of three methods have been continuously improved. All methods perform the best in terms of kappa and F1-score for the Fangchenggang dataset when the number of superpixels is 4000.

It is worth mentioning that the accuracy values achieve steady state when the number of superpixels further increases to 5000. In this case, the smaller superpixels will be not beneficial to the segmentation results. The reason is the area of each superpixel becomes so small that its shape will be

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TABLE V
COMPARISON OF THE OVERALL SEGMENTATION PERFORMANCE ON GAOFEN DATESET

| | SegNet | Deeplab V3+ | PILSG | SLIC-GGDG | DSSG |
|---|---|---|---|---|---|
| OP (%) | 72.46±0.13 | 78.24±0.34 | 79.64±0.23 | 77.86±0.31 | **81.14±0.21** |
| OA (%) | 72.53±0.33 | 78.36±0.35 | 79.35±0.31 | 77.68±0.27 | **81.26±0.20** |
| F1-score (%) | 78.36±0.14 | 83.41±0.26 | 84.67±0.17 | 82.97±0.16 | **87.93±0.13** |
| Kappa | 0.692±0.011 | 0.735±0.024 | 0.748±0.013 | 0.738±0.013 | **0.762±0.009** |

TABLE VI
TIME COSTS (S) OF DIFFERENT METHODS FOR DIFFERENT DATASETS

| Method | Simulated Image with Carton texture | Simulated Image with Shape texture | Fangchenggang Data | Poyang lake Data | Gaofen Data |
|---|---|---|---|---|---|
| SegNet | 86.91 | 86.54 | 302.67 | 311.87 | 88.69 |
| Deeplab V3+ | 123.12 | 121.63 | 443.86 | 466.23 | 113.56 |
| PILSG | 3.92 | 3.65 | 12.96 | 12.85 | 3.56 |
| SLIC-GGDG | 13.24 | 13.05 | 46.71 | 46.44 | 11.95 |
| DSSG | **1.31** | **1.33** | **5.21** | **5.17** | **1.43** |

easily affected by the speckle noise. Furthermore, to reduce the size of superpixels, the graph model needs to have more nodes to represent the superpixels. The corresponding graph convolutional network will have more parameters. This requires the device to have more memory and running time. Therefore, we set the initial superpixel size of DSSG to $16 \times 16$ in the experimental section. For various superpixel sizes, the kappa and F1-score values of DSSG remain higher than other methods, proving that our DSSG is quite robust to different sizes of superpixels. In addition, three methods have a similar trend in Figs. 14–16. The main advantage of superpixel-wise segmentation is that the number of superpixels is far less than that of pixels, which can expedite the subsequent image processing. Too small superpixels will make the superpixel-wise segmentation method lose this advantage. Hence it is not recommendatory to produce too many superpixels.

### G. Discussion on the Efficiency of Different Methods

Table VI counts the mean running time of various algorithms to segment an image of four datasets. The software configuration used for the experiments is Ubuntu 16.04 system, python 2.7, and Pytorch 0.4.1. The computer hardware configuration is 16G memory, Intel(R) Core(TM) i5-1035G4 CPU at 1.10 GHz 1.50 GHz, and NVIDIA GeForce RTX 2080 Ti graphics.

For the proposed DSSG, the required running time includes LGRP feature extraction, superpixel generation, and graph-based segmentation. As shown in Table VI, for the Fangchenggang data with size $1024 \times 1024$, the running time of our DSSG method is about 5 s, whereas the time costs of PILSG and SLIC-GGDG are 13 and 47 s, respectively. As our method can be implemented using a simple deep network, it has the highest computation efficiency, much faster than the PILSG and SLIC-GGDG methods. As pixel-wise encode–decode structure network, VGG16 used in SegNet has 16 layers and ResNet101 network in Deeplab V3+ has 101 layers. Therefore, SegNet and Deeplab V3+ have heavy computational

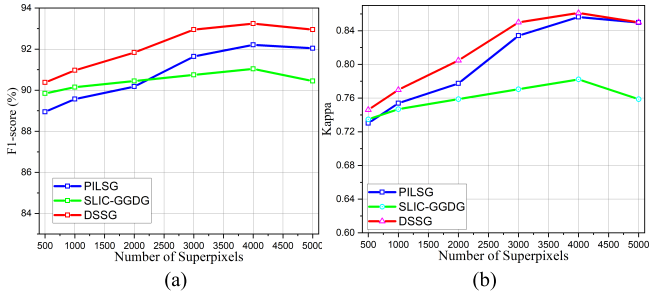



Fig. 15. F1-score and kappa curves with different superpixel numbers for the Poyang Lake dataset. (a) F1-score curves for the Poyang lake dataset. (b) Kappa curves for the Poyang lake dataset.

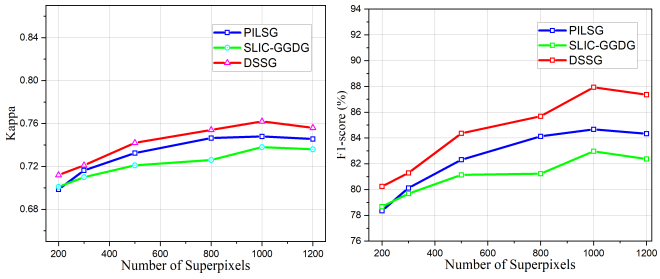



Fig. 16. F1-score and kappa curves with different superpixel numbers for the Gaofen dataset. (a) F1-score curves for the Gaofen dataset. (b) Kappa curves for the Gaofen dataset.

burdens. Their running time is more than 300 and 400 s, respectively, far longer than those of other three methods. This demonstrates that superpixels could greatly reduce the computational complexity of subsequent segmentation steps. Analyzing the average testing time of different methods on the other two real datasets, we can get similar conclusions.

The number of superpixels determines the computational complexity of our segmentation algorithm to a certain extent. In the future research, we will improve the superpixel initialization stage to enable it to adaptively change the sizes of superpixels according to the texture and boundaries of the SAR images, thereby reducing the number of superpixels. Moreover, we can also introduce sparse representation to reduce

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the occupancy of hardware resources such as memory during the running of our program, to achieve real-time segmentation.

## V. Conclusion

This article has presented a trainable end-to-end superpixel-wise segmentation network for single-polarization SAR images. First, we propose a differentiable boundary-ware clustering method instead of traditional local clustering to generate superpixels. It can leverage deep networks for estimating task-specific superpixels. Furthermore, we add a novel boundary-ware loss function for this network to suppress the interference from multiplicative speckle noise and overcome the data imbalance issue. Then, we propose a Soft-GCN to segment the whole scene SAR images based on the superpixel generation results. The superpixel generation part and graph convolution part can be trained under a unified framework to capture the global optimum parameters. The advantage of this method is that the superpixel prediction network can gradually adjust the shapes of superpixels according to the segmentation task. Experiments on simulated and real SAR datasets show that the proposed superpixel generation model can better preserve the boundary information and has good robustness to speckle. The whole segmentation method is computationally efficient and consistently achieve the state-of-the-art performance with good generalizability. In future research, we will further improve this method to perform end-to-end superpixel-wise segmentation for polarimetric SAR data.

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